Cycle Gravity Spectra and a Uniform Counterexample Family for Graffiti 197

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Abstract. Roucairol and Cazenave exhibited the cycle on 17 vertices as a counterexample to Graffiti Conjecture 197 and conjectured that every cycle of order $4r+1 \geq 17$ is likewise a counterexample, under the usual spectral-range convention. We establish a stronger uniform conclusion: every cycle C_n with $n \geq 15$ violates the proposed inequality. The proof decomposes a shifted reciprocal-distance matrix into an explicit positive-semidefinite sum of cyclic interval Gram matrices, yielding the uniform bound $\lambda_{\min}(R_n) \geq -3/2$. A harmonic-number comparison completes both parity cases without asymptotics. Exact rational interval certificates verify that the threshold is sharp within the class of cycles: $C_3, \ldots, C_{14}$ do not violate the inequality. We supply an independently executable verifier, including exact finite checks and independent Fourier spectral tests. Attribution and historical priority are separated from mathematical correctness pending independent review.

Keywords. spectral graph theory; Graffiti conjectures; reciprocal-distance matrix; circulant matrix; positive semidefinite decomposition; certified computation.

1 The problem and its conventions

The Graffiti programme proposed many inequalities connecting graph invariants; their formulation and automated falsification have a long history [1, 2, 3]. Roucairol and Cazenave [5, §5.1] reported C_{17} as a counterexample to Graffiti 197, and also checked C_{21} and C_{25}. They conjectured that every C_{4r+1} of order at least 17 is a counterexample. Their discussion emphasizes a genuine terminological ambiguity: *range* can mean a numerical span or the number of distinct values. Our claims use *only* the numerical span, as in their counterexample.

Let C_n ($n \geq 3$) have vertex set $\mathbb{Z}/n\mathbb{Z}$, ordinary graph distance $d(u,v) = \min(|u-v|, n-|u-v|)$, and adjacency matrix A_n. If $\lambda_1(A_n) \geq \cdots \geq \lambda_n(A_n)$, write $\lambda_{n-1}(A_n)$ for its second-smallest eigenvalue. Following the gravity convention reproduced in [5, §5.2], every cycle vertex has degree two and hence

$$(\mathrm{Gr}_n)_{uv} = \begin{cases} 0 & u = v, \\ \dfrac{4}{(n-1)d(u,v)} & u \neq v. \end{cases} \tag{1}$$

Let R_n denote the reciprocal-distance matrix, so $\mathrm{Gr}_n = 4R_n/(n-1)$. Define $\mathrm{spr}(M) = \lambda_{\max}(M) - \lambda_{\min}(M)$ for symmetric M. In this convention Graffiti 197 asks for

$$-\lambda_{n-1}(A_n) \leq \mathrm{spr}(\mathrm{Gr}_n). \tag{2}$$

We show that its strict reverse holds on every sufficiently large cycle, with an exact threshold.

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Theorem 1 (Infinite counterexample family)**.** *For every integer $n \geq 15$,*

$$-\lambda_{n-1}(A_n) > \mathrm{spr}(\mathrm{Gr}_n). \tag{3}$$

Thus the Roucairol–Cazenave conjecture holds, with a stronger conclusion covering all residue classes of n.

The spectral estimate driving the proof concerns the Harary or reciprocal-distance matrix, an established graph invariant [4].

2 A positive-semidefinite identity

Let $m = \lfloor n/2 \rfloor$, let J be the all-ones matrix, and, for $2 \leq j \leq m$, let B_j be the $n \times n$ binary circulant matrix whose vth column is the indicator of the cyclic interval $\{v, v+1, \ldots, v+j-1\}$. Set

$$w_j = \begin{cases} \dfrac{2}{(j-1)j(j+1)} & 2 \leq j < m, \\ \dfrac{1}{m(m-1)} & j = m. \end{cases} \tag{4}$$

All these coefficients are strictly positive.

Lemma 1 (Exact Gram decomposition)**.** *For every $n \geq 4$,*

$$R_n + \frac{3}{2}I = \frac{1}{m}J + \sum_{j=2}^{m} w_j B_j B_j^{\mathsf{T}}. \tag{5}$$

In particular $\lambda_{\min}(R_n) \geq -3/2$.

Proof. Two cyclic intervals of length $j \leq m$ at distance $d \leq m$ share precisely $(j-d)_+ = \max(j-d, 0)$ vertices. Hence the (u,v) entry of $B_j B_j^{\mathsf{T}}$ equals $(j - d(u,v))_+$. For $1 \leq d \leq m$, the discrete convex interpolation of $f_d = 1/d - 1/m$ satisfies

$$\frac{1}{d} - \frac{1}{m} = \sum_{j=d+1}^{m} w_j (j-d),$$

with the coefficients in (4): for $j < m$ they are $1/(j-1) - 2/j + 1/(j+1)$, and the terminal coefficient is $1/(m-1) - 1/m$. Thus the off-diagonal entries in (5) equal $1/d$. On the diagonal,

$$\frac{1}{m} + \sum_{j=2}^{m} j w_j = \frac{1}{m} + \sum_{j=2}^{m-1} \left(\frac{1}{j-1} - \frac{1}{j+1} \right) + \frac{1}{m-1} = \frac{3}{2},$$

including the case $m = 2$ (empty middle sum). This proves the matrix identity. Both J and each $B_j B_j^{\mathsf{T}}$ are positive semidefinite, giving the eigenvalue bound. □

The identity is finite-dimensional and exact; its verification requires no numerical eigensolver. Let $H_m = \sum_{j=1}^{m} j^{-1}$. As R_n is nonnegative with constant row sums, its largest eigenvalue is that row sum. In particular,

$$\lambda_{\max}(R_n) = \begin{cases} 2H_m & n = 2m+1, \\ 2H_{m-1} + 1/m & n = 2m. \end{cases} \tag{6}$$

Combining (1), Lemma 1 and (6) yields

$$\operatorname{spr}(\mathrm{Gr}_n) \le \frac{4}{n-1} \left(\lambda_{\max}(R_n) + \frac{3}{2} \right). \tag{7}$$

3 Proof of the universal statement

For cycles the adjacency spectrum is exactly

$$\{ 2\cos(2\pi k/n) : k = 0, 1, \dots, n-1 \}.$$

Ordering these eigenvalues, with multiplicity, gives

$$-\lambda_{n-1}(A_n) = \begin{cases} 2\cos(\pi/(2m+1)) & n = 2m+1, \\ 2\cos(\pi/m) & n = 2m. \end{cases} \tag{8}$$

Proof of Theorem 1. For odd $n = 2m+1$ ($m \ge 7$), (7) bounds the gravity spread by

$$q_o(m) = \frac{4H_m + 3}{m}.$$

This is strictly decreasing: $q_o(m) > q_o(m+1)$ follows from $4H_m + 3 > 4m/(m+1)$. Meanwhile $2\cos(\pi/(2m+1))$ is increasing in m. It therefore suffices to check $m = 7$. Using $H_7 = 363/140$, $\cos x \ge 1 - x^2/2$, and the rational inequality $\pi < 22/7$, we get

$$2\cos(\pi/15) - q_o(7) > \left[2 - \left(\frac{22}{105} \right)^2 \right] - \frac{468}{245} = \frac{506}{11025} > 0.$$

This proves the odd case.

For even $n = 2m$ ($m \ge 8$), define

$$S_m = 2H_{m-1} + \frac{1}{m} + \frac{3}{2}, \qquad q_e(m) = \frac{4S_m}{2m-1}.$$

Now $S_{m+1} - S_m = 1/m + 1/(m+1)$; thus $(2m-1)(S_{m+1} - S_m) < 4 < 2S_m$ for $m \ge 2$, proving $q_e(m) > q_e(m+1)$. The quantity $2\cos(\pi/m)$ increases. At $m = 8$,

$$2\cos(\pi/8) - q_e(8) > \left[2 - \left(\frac{11}{28} \right)^2 \right] - \frac{1907}{1050} = \frac{1733}{58800} > 0.$$

Both estimates compare the left-hand side of (3) to an *upper bound* on its right-hand side. The strict inequality follows for all $n \ge 15$. □

Corollary 1. *For every $r \ge 4$, the cycle C_{4r+1} refutes Graffiti 197 in the numerical-span convention, as conjectured in [5].*

4 Sharpness and certificate design

To check the finite complement without trusting floating-point eigenvalues, use a Fourier eigenvalue of R_n at frequency $k = \lfloor n/2 \rfloor$:

$$\eta_k = 2 \sum_{d=1}^{\lfloor (n-1)/2 \rfloor} \frac{\cos(2\pi k d/n)}{d} + \mathbf{1}_{2 \mid n} \frac{(-1)^k}{n/2}. \tag{9}$$

We have $\lambda_{\min}(R_n) \le \eta_k$, regardless of whether this mode is globally minimal. It follows that

$$-\lambda_{n-1}(A_n) - \operatorname{spr}(\mathrm{Gr}_n) \le 2\cos\theta_n - \frac{4}{n-1}(s_n - \eta_k), \tag{10}$$

where $s_n = \lambda_{\max}(R_n)$ and $\theta_n = \pi/n$ for odd n, $2\pi/n$ for even n.

The accompanying Python verifier encloses π by exact rational alternating-series estimates from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Every cosine is enclosed by exact rational evaluations of its Taylor polynomial of degree 24, with the explicit remainder bound $(22/7)^{25}/25!$. It then checks that the *upper* bound on the right-hand side of (10) is strictly negative for each $n = 3, \dots, 14$. All acceptance-path comparisons are between exact Python rational numbers. Thus none of these twelve smaller cycles refutes (2).

Theorem 2 (Sharp cycle threshold, computer-assisted finite part)**.** *Among simple cycle graphs C_n ($n \ge 3$), violation of (2) occurs if and only if $n \ge 15$. The positive direction has the analytic proof above; the finite converse is certified by the rational interval algorithm just described.*

5 Formal certification: five-stage proof ledger

A pinned Lean 4/mathlib v4.24.0 project certifies proof-critical identities of Lemma 1; sources and lockfile reside in the public Newton Astro Labs repository. The full graph-spectral theorem is *not yet kernel-certified.*

Stage 1: rational margins and Gram positivity. `NewtonConjectures.lean` certifies the strict base-case margins 506/11025 and 1733/58800, rational reciprocal second differences, positive weights, and nonnegative weighted Gram quadratic forms.

Stage 2: universal reciprocal-distance kernel. `CycleKernel.lean` proves telescoping for all $f : \mathbb{N} \to \mathbb{Q}$. For $1 \le d < m$, the reciprocal specialization is

$$\frac{1}{d} = \frac{1}{m} + \sum_{j=d+1}^{m-1} (j-d)\frac{2}{(j-1)j(j+1)} + \frac{m-d}{m(m-1)}. \tag{11}$$

Stage 3: exact cyclic Gram entries. `CyclicIntersection.lean`, `CyclicZMod.lean` and `CyclicGram.lean` formalize genuine cyclic incidence matrices over $\mathbb{Z}/n\mathbb{Z}$ and their `Matrix.mul`. For all vertices u, v and $2j \le n$, including wraparound, disjoint arcs, and antipodes, the kernel proves

$$(B_j B_j^{\mathsf{T}})_{uv} = \max\{j - d(u,v), 0\}. \tag{12}$$

Stage 4: actual weighted Gram diagonal. For every $m \ge 2$, `GramDiagonal.lean` certifies

$$\frac{1}{m} + \sum_{j=2}^{m-1} \frac{2}{(j-1)(j+1)} + \frac{1}{m-1} = \frac{3}{2}. \tag{13}$$

It also proves the identical diagonal for the *actual weighted incidence matrices* of every C_n, $n \ge 4$.

Stage 5: all off-diagonal distances. For every $m \ge 2$ and $1 \le d \le m$, `OffDiagonal.lean` proves

$$\frac{1}{m} + \sum_{j=2}^{m} w_j \max\{j-d, 0\} = \frac{1}{d}. \tag{14}$$

These are exactly the weights of (4), with the terminal case $d = m$ included using natural truncated subtraction.

Independent verification. GitHub Actions run 39131413230 passed 7,358 Lean build jobs. Its axiom audit accepted 100 declarations with only `propext`, `Classical.choice`, `Quot.sound`; the proof-placeholder check passed. Exact `fractions.Fraction` arithmetic independently confirmed all 7,259 cases $2 \le m \le 120$, $1 \le d \le m$. See the versioned proof ledger.

Claim boundary. These certificates supply the entries of (5) but do not yet include a *single* kernel-certified all-entry matrix theorem, the complete cycle-specific spectral bound, adjacency eigenvalue ordering, two harmonic comparisons, or the formal finite converse $C_3, \ldots, C_{14}$. The paper's analytic argument and exact finite certificate remain separately stated; priority and specialist review remain open.

6 Reproducibility and claim boundaries

The paper's proof is self-contained apart from classical elementary facts ($\pi < 22/7$, $\cos x \ge 1 - x^2/2$, and the cycle adjacency spectrum). The accompanying `verify.py` provides four checks: rational matrix-identity entries for orders 4 to 40, rational bounds for all small orders 3 to 14, rational evaluation of both boundary inequalities, and an independent floating-point Fourier cross-check through order 160. The numerical cross-check is deliberately *not* an ingredient of the theorem. Proof-critical component lemmas have now been independently kernel-checked in Lean 4/mathlib as documented in Section 5; the complete graph-theoretic theorem has not been formally certified.

Graffiti 197 itself was already refuted by Roucairol and Cazenave [5]; its resolution is not claimed as a new discovery. The contribution under review is the uniform analytic infinite-family theorem, the extended threshold, and the exact Gram decomposition. The targeted literature audit did not identify an earlier proof of this specific strengthening, but that negative search result is not a priority certificate. Mathematical validity, originality, and suitability for publication remain open to independent assessment.

References

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